

Zerun Niu

MPhil Student in Computer Science

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Summary

MPhil student at The University of Sydney working on federated learning, semantic communication, and efficient AI systems. Research interests include distributed optimization, edge intelligence, parameter-efficient tuning, and ML systems for resource-constrained environments.

Education

Mas- ter of	The University of Sydney , Computer Science	Sydney, Australia
Phi- loso- phy	• Supervisor: Nguyen Tran • Research focus: Federated Learning, Semantic Communication, and Efficient AI Systems	2025 – present
Bach- elor of	The University of Sydney , Data Science	Sydney, Australia
Advanced Com- put- ing	• Final years WAM: 77/100 • Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Distributed Systems	2025

Experience

DUAL Group , The University of Sydney , Research Assistant	Sydney, Australia
Research engineering and experimentation for distributed learning and efficient AI systems. • Built reproducible training and evaluation pipelines with configs, seeds, logging, and automated reporting. • Implemented baselines, targeted ablations, and publication-quality figures and tables for paper submissions. • Addressed reviewer feedback through additional experiments and clarity edits with supervisors and co-authors. • Coordinated experiment ownership, timelines, and artifact handoff across code, plots, and tables. • Designed and deployed the official DUAL Group website.	Apr 2024 – present 2 years 3 months

Publications

Distributionally Robust Wireless Semantic Communication with Large AI Models	2025
L. T. Le, S. H. Wanasekara, <i>Zerun Niu</i> , et al. 10.48550/arXiv.2506.03167 (IEEE Journal on Selected Areas in Communications)	
Federated Deep Equilibrium Learning over Resource-Constrained Edge Networks	2025
L. T. Le, <i>Zerun Niu</i> , T. D. Nguyen, et al. (Submitted to IEEE Internet of Things Journal)	

Projects

Adaptive Low-Rank Optimization for Federated Learning

2025

Undergraduate thesis integrating DyLoRA into a federated learning pipeline.

- Reduced communication cost by 80% without loss of accuracy.
- Improved training stability across clients with reproducible PyTorch experiments.
- Result: 85/100, High Distinction.

DUAL Group Website

2024-2025

Designed and deployed the official website for the Distributed compUting, optimizAtion, and Learning group.

- Introduced group members, publications, and research projects.
- Website: <https://dual-grp.github.io/website-dual/>

Teaching Readiness

Computer Science and Data Science Teaching Support

Prepared to support labs, tutorials, code reviews, and project assessment.

- Subjects: Algorithms and Data Structures, Computer Systems, Machine Learning, Deep Learning, NLP, Data Science Foundations, Distributed Systems.
- Assessment: assignments, coding practicals, project reports, oral demos, rubric-aligned marking.
- Student support: consultations, Ed discussion, code reviews, debugging practices, academic integrity and citation support.

Skills

GenAI and Multimodal: LLM/VLM orchestration, agents, tool use, multimodal adapters, LoRA, alignment, safety guardrails, jailbreak defense, RAG, evaluation.

Efficient Tuning: LoRA, DyLoRA, PEFT workflows, quantization, distillation.

ML Engineering: PyTorch, Hugging Face Transformers, PEFT, scikit-learn, reproducible training pipelines, experiment tracking.

Distributed and Data Systems: Ray, Dask, Databricks, Spark, Hadoop, Airflow, FastAPI, tracing, observability.

Programming: Python, SQL, Java, C/C++, Git/GitHub, Linux, Jupyter, API development.

Awards

Undergraduate Thesis High Distinction

2025

Adaptive Low-Rank Optimization for Federated Learning, score 85/100.